

Monir Hosen

Fazlul Huq Muslim Hall, University of Dhaka, Dhaka-1000, Bangladesh
monirakash2002@gmail.com | +880 1940237160
Github | LinkedIn | Kaggle

Detail-oriented Business Intelligence Analyst with a strong statistics foundation and hands-on expertise in SQL, Python, R, Power BI, and Tableau and problem solving. Passionate about turning healthcare data into insights that improve patient outcomes and operational efficiency. Currently pursuing a master's degree while applying machine learning to solve real-world healthcare challenges.

Education

Master of Science (with Thesis) in Computer Science and Engineering 2025 – Present
University of Dhaka

Bachelor of Science in Statistics 2020 – 2025
University of Dhaka CGPA: 3.28 out of 4

Skills

Query & Programming Languages: SQL, Python, R Programming

BI & Data Visualization: Power BI, Tableau, Excel (Pivot Tables, Pivot Charts, Slicers, KPI Dashboards)

Databases: MySQL, PostgreSQL

Data Analysis & Reporting: EDA, Data Cleaning, Data Mining, Business Requirement Documentation

Machine Learning & Deep Learning: ML (Supervised/Unsupervised), CNNs, RNNs, LSTMs

Frameworks & Libraries: Numpy, Pandas, Matplotlib, scikit-learn, scikit-image, PyTorch

Projects

1. Customer Shopping Behavior Analytics | *Python* | *PostgreSQL* | *SQL* | [GitHub]
Power BI

- Built an end-to-end business intelligence solution, transforming raw customer shopping data into an interactive analytics dashboard for stakeholder reporting.
- Developed analytical SQL queries using **CTEs, window functions, ranking, subqueries, and aggregations** to solve **18** real-world business problems related to customer behavior, revenue, subscriptions, discounts, and seasonal demand.

- Performed **customer segmentation** (New, Returning, Loyal) and implemented an RFM-lite customer value analysis using purchase history and spending patterns.
- Designed an interactive Power BI dashboard featuring **KPI cards, revenue trends, customer demographics, product performance, subscription insights, and dynamic filtering.**
- **Automated the analytics workflow** from Python-based data cleaning through SQL analysis to executive-level dashboard reporting.

2. Aroga Medicine E-commerce Data Pipeline | *Python* | *PostgreSQL* | *SQL* | *XPath* [GitHub]

- Built an end-to-end data pipeline that scrapes a live e-commerce site with **50,000+** products and turns it into a clean, query-ready PostgreSQL database.
- Designed a star-schema database (**7 tables**) with dimension and fact tables for brands, generics, categories, and daily price/stock snapshots, enabling time-series pricing and stockout analysis.
- Wrote a resumable Python scraper using XPath to extract **20+** structured fields per product, including price, discount, stock status, and prescription requirements.
- Validated data integrity with automated checks for primary keys, foreign keys, and duplicate records; found and fixed **a real data-truncation bug affecting 208 records** before loading into production.

3. Mango Varieties Classification Using a Custom CNN Model [GitHub]

- Developed a custom Convolutional Neural Network to classify multiple mango varieties from image data.
- Implemented model training, validation, and performance monitoring using metrics such as accuracy, precision, recall, and confusion matrix analysis.

Research Publications

Impact of Socioeconomic, Demographic, Maternal, Infant and Healthcare Factors on Early Initiation of Breastfeeding in Bangladesh: Evidence from Bangladesh Demographic Health Survey (BDHS) 2022 Data [Link]

PLOS ONE, 2026

- **Co-authored a peer-reviewed** research article analyzing nationally representative **BDHS 2022** data.
- Applied statistical methods to identify socioeconomic, demographic, maternal, infant, and **healthcare factors** associated with **early initiation of breastfeeding** in Bangladesh.

Standardized Test Scores

- **IELTS:** Listening 7 | Reading 6.5 | Writing 6 | Speaking 6 | Overall 6.5